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Forensics Walkthrough: TryHackMe



Sakshi Aggarwal · [Follow](#)

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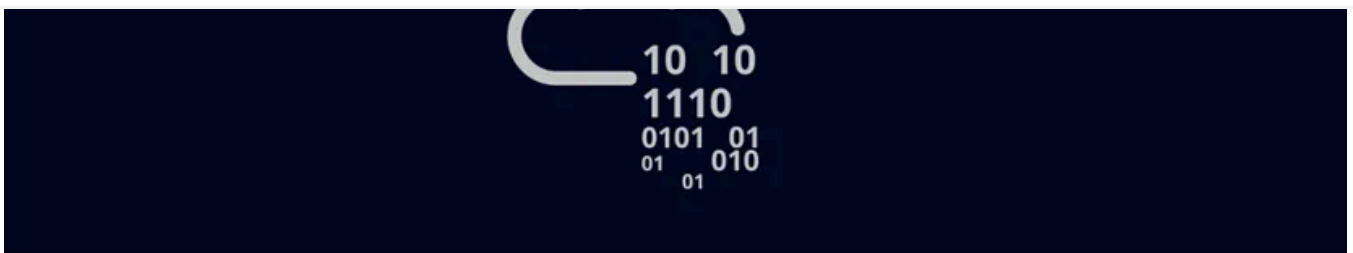
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This is a memory dump of compromised system, do some forensics kung-fu to explore the inside.

Task [1]: Volatility forensics

#1 Download the victim.zip

Ans. No answer needed

After downloading the file , launch the **Volatility (memory forensics tool)** and type the command **volatility -h** to get the help menu and find the plugins to answer the questions.

#2 What is the OS of this Dump? (Just write OS name in small)

Get the information of the image by using the plugin **imageinfo**. Look at the suggested profiles, it seems that the victim was using windows OS.

Ans. windows

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw imageinfo
Volatility Foundation Volatility Framework 2.6
INFO : volatility.debug : Determining profile based on KDBG search...
      Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_23418, Win2008R2SP1x64, Win7SP1x64_23418
      AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
      AS Layer2 : FileAddressSpace (/home/sakshi/Downloads/volatility_2.6_lin64_standalone/victim.raw)
      PAE type : No PAE
      DTB : 0x187000L
      KDBG : 0xf800028420a0L
      Number of Processors : 1
      Image Type (Service Pack) : 1
      KPCR for CPU 0 : 0xfffff80002843d00L
      KUSER_SHARED_DATA : 0xfffff78000000000L
      Image date and time : 2019-05-02 18:11:45 UTC+0000
      Image local date and time : 2019-05-02 11:11:45 -0700
```

#3 Whats is the PID of SearchIndexer ?

Use **pslist** plugin to list out all the processes that were running at the time of image creation.

Ans. 2180

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw --profile=Win7SP1x64
pslist
```

0xfffffa8003367060	SearchIndexer.	2180	504	11	629	0	0	2019-05-02 18:03:32 UTC+0000
--------------------	----------------	------	-----	----	-----	---	---	------------------------------

(A blue circle highlights the PID 2180, with an arrow pointing to it and the label "PID")

#4 What is the last directory accessed by the user? (Just write last folder name as it is?)

Shellbags plugin command can be used to gather all the information about a viewed directory such as size, position, and icon. Using this command, we are able to track down the folder accessed by the user. Check the access date and time and get the name of last directory accessed.

Ans. deleted_files

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw --profile=Win7SP1x64
shellbags
Volatility Foundation Volatility Framework 2.6
Scanning for registries....
Gathering shellbag items and building path tree...
*****
Registry: \??\C:\Users\victim\ntuser.dat
Key: Software\Microsoft\Windows\Shell\Bags\1\Desktop
Last updated: 2019-05-02 07:00:41 UTC+0000
Value      File Attr      File Name      Modified Date      Create Date      Access Date
-----
ItemPos1366x664x96(1)  Firefox.lnk  2019-04-11 13:29:10 UTC+0000  2019-04-11 13:29:10 UTC+0000  2019-04-11 13:29:10 UTC
+0000 ARC          Firefox.lnk
ItemPos1366x664x96(1)  HxD.lnk      2019-04-13 06:00:40 UTC+0000  2019-04-13 06:00:40 UTC+0000  2019-04-13 06:00:40 UTC
+0000 ARC          HxD.lnk
ItemPos1366x664x96(1)  WIRESH-1.LNK 2019-04-13 06:31:06 UTC+0000  2019-04-13 06:31:06 UTC+0000  2019-04-13 06:31:06 UTC
+0000 ARC          Wireshark.lnk
ItemPos1366x664x96(1)  LIFEVI-1     2019-04-23 06:14:40 UTC+0000  2019-04-23 07:46:36 UTC+0000  2019-04-23 06:14:40 UTC
+0000 DIR          Life Virus Samples
ItemPos1366x664x96(1)  NEWFOL-1     2019-04-13 08:02:44 UTC+0000  2019-04-13 08:02:44 UTC+0000  2019-04-13 08:02:44 UTC
+0000 DIR          New folder
ItemPos1366x664x96(1)  ~res-x64.txt 2019-04-13 08:19:32 UTC+0000  2019-04-13 08:19:32 UTC+0000  2019-04-13 08:19:32 UTC
+0000 ARC          ~res-x64.txt
ItemPos1366x664x96(1)  ~RES-X-1.TXT 2019-04-27 10:37:32 UTC+0000  2019-04-27 10:37:32 UTC+0000  2019-04-27 10:37:32 UTC
+0000 ARC          ~res-x64_0000.txt
ItemPos1366x664x96(1)  ANALYS-1.TXT 2019-04-18 01:08:50 UTC+0000  2019-04-18 00:57:28 UTC+0000  2019-04-18 00:57:28 UTC
+0000 ARC          Analysis Deatils.txt
ItemPos1366x664x96(1)  DEPEND-1.LNK 2019-04-13 07:31:04 UTC+0000  2019-04-13 07:31:04 UTC+0000  2019-04-13 07:31:04 UTC
+0000 ARC          depends - Shortcut.lnk
ItemPos1366x664x96(1)  emotet.txt   2019-04-27 10:30:10 UTC+0000  2019-04-27 10:30:10 UTC+0000  2019-04-27 10:30:10 UTC
+0000 ARC          emotet.txt
ItemPos1366x664x96(1)  EMOTET-1.HIV 2019-04-27 10:31:40 UTC+0000  2019-04-27 10:31:28 UTC+0000  2019-04-27 10:31:28 UTC
+0000 ARC          emotet_reschot.hiv
```

```
File Actions Edit View Help
C:\Program Files (x86)\Capture
*****
Registry: \??\C:\Users\victim\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\1\2\0
Last updated: 2019-04-27 10:48:33 UTC+0000
Value      Mru      File Name      Modified Date      Create Date      Access Date      File
Attr      Path
-----
0          0          deleted_files  2019-04-27 10:30:26 UTC+0000  2019-04-27 10:38:24 UTC+0000  2019-04-27 10:38:24 UTC+0000  NI,
DIR      Z:\logs\deleted_files
*****
Registry: \??\C:\Users\victim\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\5\2\2
Last updated: 2019-04-23 06:14:25 UTC+0000
Value      Mru      File Name      Modified Date      Create Date      Access Date      File
Attr      Path
```

Task [2]: Task2

#1 There are many suspicious open port, which is it?(protocol:port)

connections or connscan plugins are available for scanning open port but these are restricted to Windows XP and 2003. So, we can use **netscan** plugin.

Ans. UDP:5005

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw --profile=Win7SP1x64
netscan
Volatility Foundation Volatility Framework 2.6
Offset(P)      Proto      Local Address      Foreign Address      State      Pid      Owner      Creat
ed
0x5c201ca0      UDPv4      0.0.0.0:5005      *:5005               *:*        2464      wmpnetwk.exe  2019-
05-02 18:05:14 UTC+0000
```


#2 Vads tag and execute protection are strong indicators of malicious processes, can you find which are they? (Pid1;Pid2;Pid3...)

malfind is the plugin that can be used to find out the hidden and injected code(malicious processes). When we run the command we found out three processes that are malicious.

Ans. 1860;1820;2464

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw --profile=Win7SP1x64
malfind
Volatility Foundation Volatility Framework 2.6
Process: explorer.exe Pid: 1860 Address: 0x3ee0000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: CommitCharge: 1, MemCommit: 1, PrivateMemory: 1, Protection: 6
0x03ee0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x03ee0010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

```
Process: wmpnetwk.exe Pid: 2464 Address: 0x280000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: CommitCharge: 16, MemCommit: 1, PrivateMemory: 1, Protection: 6
0x00280000 41 ba 80 00 00 00 48 b8 38 a1 e6 ff fe 07 00 00 A.....H.8.....
0x00280010 48 ff 20 90 41 ba 81 00 00 00 48 b8 38 a1 e6 ff H...A.....H.8...
0x00280020 fe 07 00 00 48 ff 20 90 41 ba 82 00 00 00 48 b8 ....H...A.....H.
0x00280030 38 a1 e6 ff fe 07 00 00 48 ff 20 90 41 ba 83 00 8.....H...A...
```

```
Process: svchost.exe Pid: 1820 Address: 0x24f0000
Vad Tag: VadS Protection: PAGE_EXECUTE_READWRITE
Flags: CommitCharge: 128, MemCommit: 1, PrivateMemory: 1, Protection: 6
0x024f0000 20 00 00 00 e0 ff 07 00 0c 00 00 00 01 00 05 00 .....
0x024f0010 00 42 00 50 00 30 00 70 00 60 00 00 00 00 00 00 .B.P.0.p.`.....
0x024f0020 48 8b 45 28 c7 00 00 00 00 00 00 c7 40 04 00 00 00 H 5(  )
```

Task [3] IOC SAGA

IOC(Indicators of compromise) are pieces of forensic data found inside the system entries log and files. This data is then used to identify malicious activity. Since we have identified all the malicious process on the previous task, we can dump the memory of to process to identify the malicious activity.

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw -p 1860,1820,2464 --p
rofile=Win7SP1x64 memdump -D .
Volatility Foundation Volatility Framework 2.6
*****
Writing explorer.exe [ 1860] to 1860.dmp
*****
Writing svchost.exe [ 1820] to 1820.dmp
*****
Writing wmpnetwk.exe [ 2464] to 2464.dmp
*****
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw --profile=Win7SP1x64
```

By using the grep command and hint we can easily find the answer.

Command used is: `strings 1820.dmp | grep "`

strings command will get all the strings present in dump file and grep will match the strings with provided hint and give us all the matching answers.

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ strings 1820.dmp | grep '\<www\.go\.\.\.\.ru\>'
www.google.ru
www.go4win.ru
www.gocaps.ru
www.goporn.ru
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ strings 1820.dmp | grep '\<www\.i\.\.\.\.com\>'
www.ikaka.com
http://www.iciba.com/search?s=%si
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ strings 1820.dmp | grep '\<www\.ic\.\.\.\.\.com\>'
www.icsalabs.com
```

#1 'www.go****.ru' (write full url without any quotation marks)

Ans. www.goporn.ru

#2 'www.i****.com' (write full url without any quotation marks)

Ans. www.ikaka.com

#3 'www.ic*****.com'

Ans. www.icsalabs.com

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ strings 1820.dmp | grep '\<202\.\.\.\.233\.\.\.\>'
202.107.233.211
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ strings 1820.dmp | grep '\<...\.200\...\164\>'
phttp://209.200.12.164/drm/provider_license_v7.php
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ strings 1820.dmp | grep '\<209\190\.\.\.\.\.\>'
http://209.190.122.186/drm/license-savenow.asp
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$
```

#4 202.***.233.*** (Write full IP)

Ans. 202.107.233.211

#5 ***.200.**.164 (Write full IP)

Ans. 209.200.12.164

#6 209.190.*****

Ans. 209.190.122.186

#7 What is an unique environmental variable of PID 2464

envvars plugin is used to display the environment variables.

Ans. OANOCACHE

```
sakshi@kali:~/Downloads/volatility_2.6_lin64_standalone$ ./volatility_2.6_lin64_standalone -f victim.raw -p 2464 --profile=Win7SP1x64 envvars
Volatility Foundation Volatility Framework 2.6
```

Pid	Process	Block	Variable	Value
2464	wmpnetwk.exe	0x00000000002c47a0	ALLUSERSPROFILE	C:\ProgramData
2464	wmpnetwk.exe	0x00000000002c47a0	APPDATA	C:\Windows\ServiceProfiles\NetworkService\AppData\Roaming
2464	wmpnetwk.exe	0x00000000002c47a0	CommonProgramFiles	C:\Program Files\Common Files
2464	wmpnetwk.exe	0x00000000002c47a0	CommonProgramFiles(x86)	C:\Program Files (x86)\Common Files
2464	wmpnetwk.exe	0x00000000002c47a0	CommonProgramW6432	C:\Program Files\Common Files
2464	wmpnetwk.exe	0x00000000002c47a0	COMPUTERNAME	VICTIM-PC
2464	wmpnetwk.exe	0x00000000002c47a0	ComSpec	C:\Windows\system32\cmd.exe
2464	wmpnetwk.exe	0x00000000002c47a0	FP_NO_HOST_CHECK	NO
2464	wmpnetwk.exe	0x00000000002c47a0	LOCALAPPDATA	C:\Windows\ServiceProfiles\NetworkService\AppData\Local
2464	wmpnetwk.exe	0x00000000002c47a0	NUMBER_OF_PROCESSORS	1
2464	wmpnetwk.exe	0x00000000002c47a0	OANOCACHE	1
2464	wmpnetwk.exe	0x00000000002c47a0	OS	Windows_NT
2464	wmpnetwk.exe	0x00000000002c47a0	Path	C:\Windows\system32;C:\Windows;C:\Windows\Sys

Thank you!!!

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Responses (1)



What are your thoughts?

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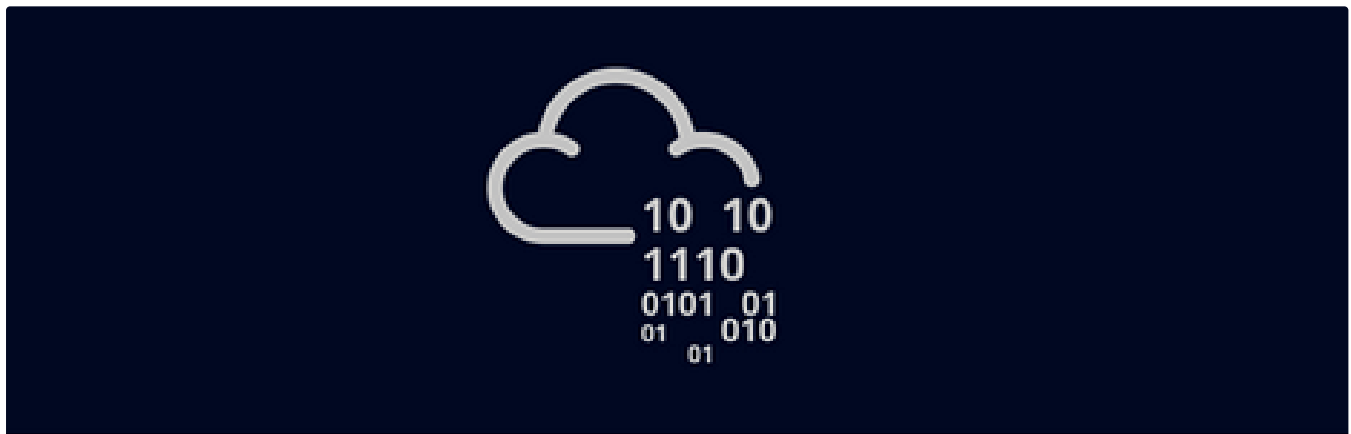
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Thanks Bro !

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**t**

ite, hash cracking and escalate privileges in this box!

**Sakshi Aggarwal**

TryHackMe: Brute It Walkthrough

Task [1]: About this box

Nov 7, 2020



11



2





Git Happens

[Start AttackBox](#)

Boss wanted me to create a prototype, so here it is! We even used something called "version contro deploying this really easy!

 Sakshi Aggarwal

TryHackMe: Git Happens Walkthrough

The only task in this room is to capture the flag. Deploy the machine.

Oct 23, 2020  12



Musical Stego

[Start AttackBox](#)[Awards](#)

This audio file is hiding some things, are you able to extract enough data to reveal the flag?

 Sakshi Aggarwal

TryHackMe: Musical Stego Walkthrough

Task [1]: Musical Steganography Challenge

Oct 13, 2020





OhSINT

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Are you able to use open source intelligence to solve this challenge?

 Sakshi Aggarwal

TryHackMe: OhSINT Walkthrough

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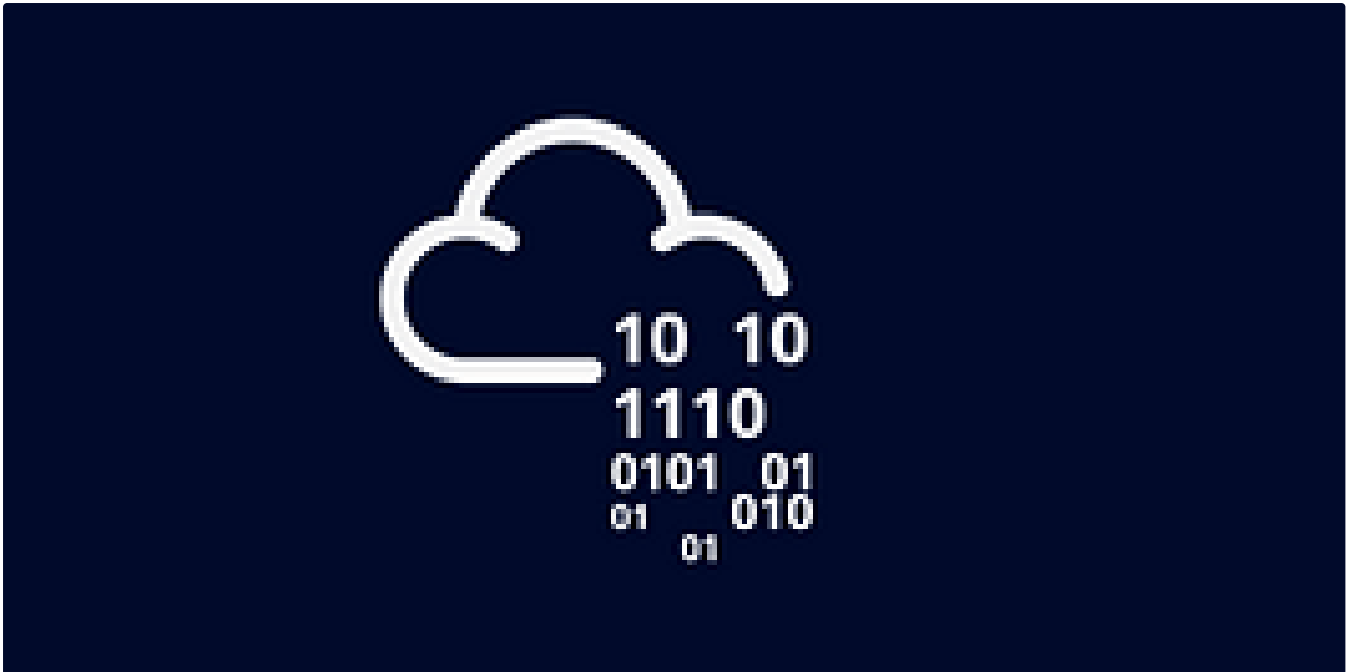


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TryHackMe | Brute Force Heroes | WriteUp

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Oct 3, 2024



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5		student3		Student3				studi
9		anatacker		Ana Tacker				
10		THM{Got.the.User}		X				
11		qweqwe		qweqwe				

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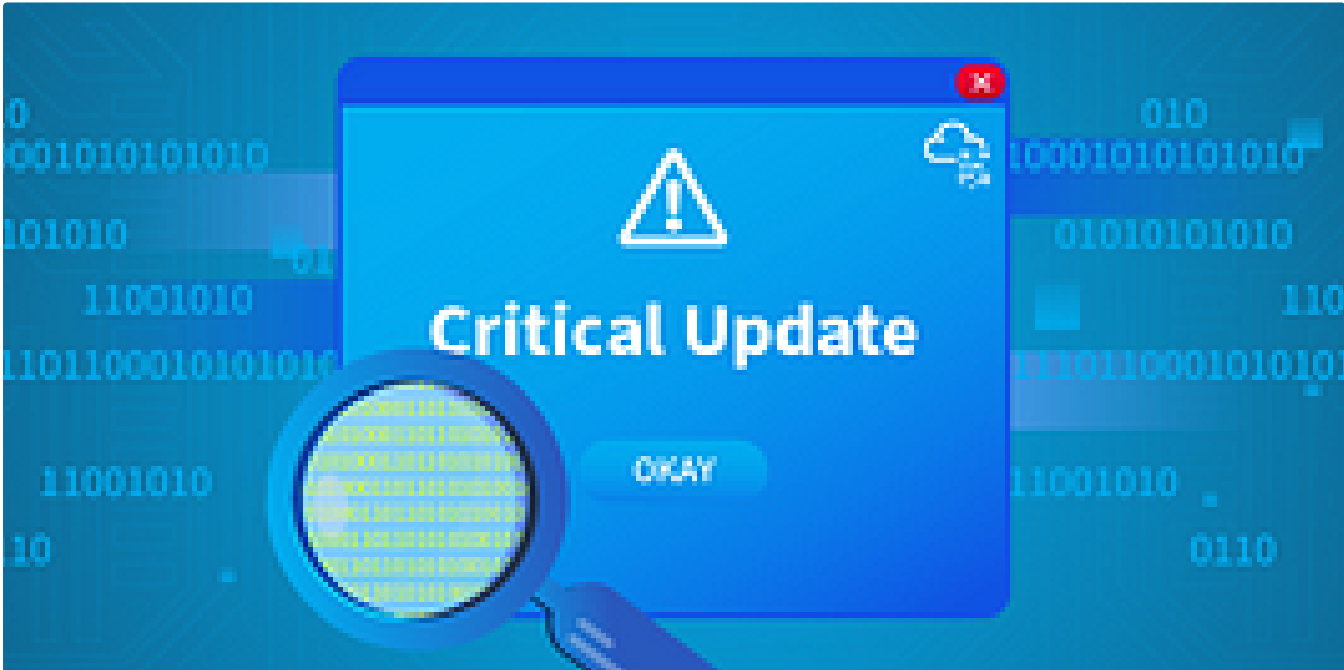
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Key points: Session Management | Authentication | Authorisation | Session Management Lifecycle | Exploit of vulnerable session management...

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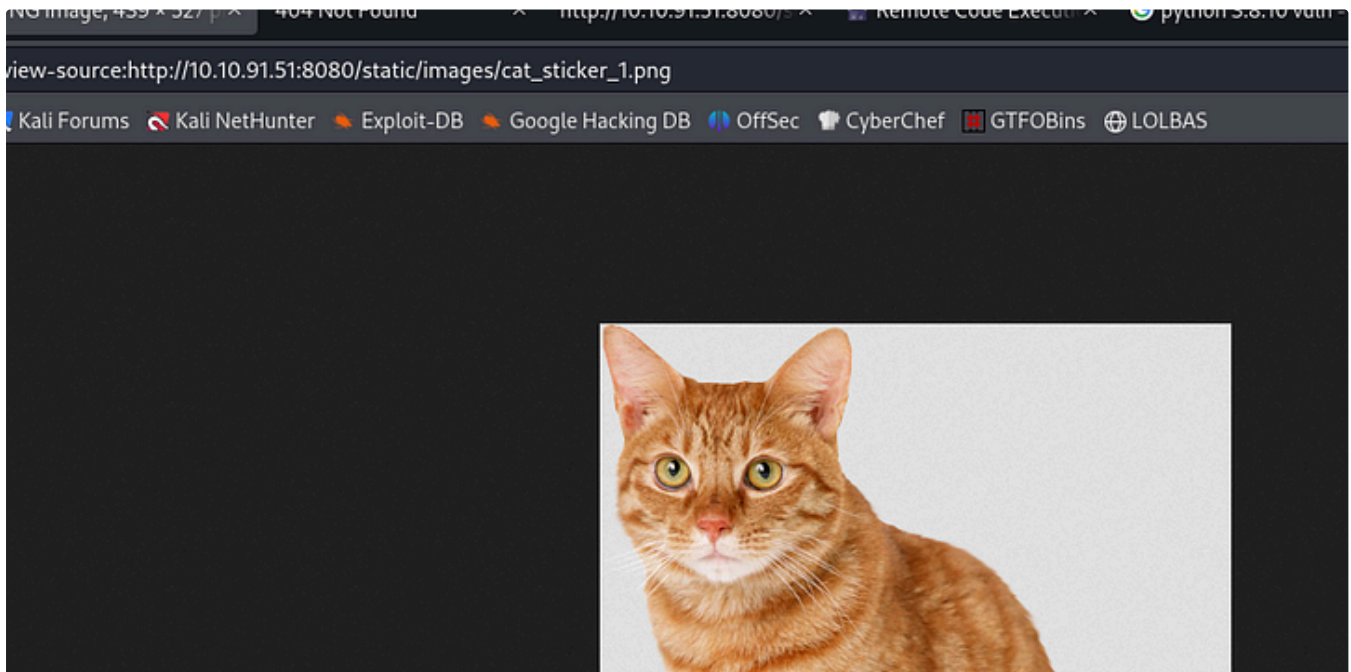


Atharva

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